Sean Yan
64322415
CPSC 320 A2

1 Slow as a logarithm

expanding on $\log_b(n) \in O(n^x)$ with the definition of big O,

$$\exists c \in \mathbb{R}, \exists n_0 \in \mathbb{R}, c > 0$$
$$\forall n \geq n_0 > 0,$$
$$\log_b(n) \leq c \cdot n^x$$

for every $b > 1$ and $x > 0$
let $n_0 = 1$ and $c = 1$

$$\log_b(1) \leq 1 \cdot 1^x$$
$$0 \leq 1 \cdot 1^x$$
$$0 \leq 1$$

so $n_0 = 1$ and $c = 1$ is an valid choosing of c and n_0

and we want, by the definition of little o

$$\lim_{n \to \infty} \left(\frac{\log_b(n)}{n^x} \right) = 0$$

Question 2: I wanna be the very best

a)

this is a combination question.

Question 3

3a)

To prove that $\ln(n!) \in O(n\ln(n))$, we choose c and n_0 and follow the definition of big O. We began with the left hand side $\ln(n!)$ and slowly expand and perform algebraic manipulations to show that we can arrive at to $c \cdot n\ln(n)$ with an inequality in the chain of those algebraic manipulations.

The critical logic to solving this proof, is to realize, that products in the natural log turns into addition. And, each of the terms in $\ln(n-i)$ for some $i \in [0, n]$, is smaller or equal to $\ln(n)$ because it's a increasing function. which just means a larger input to the natural log will yield a larger output. And we choose $n_0 = 1$, because that's where the natural log is positive. With that, we can build up the expression $n\ln(n)$.

choose $c = 1$, with no particular reason other than, it let us build up to the definition of Big O

Proof:

Choose $c = 1, n_0 = 1$

$$
\begin{aligned}
\ln(n!) &= \ln(n \cdot (n-1)!) \\
&= \ln(n) + \ln((n-1)!) \\
&= \ln(n) + \ln(n-1) + ... + \ln(1) && \text{there are } n \text{ terms} \\
&\leq \ln(n) + \ln(n) + ... + \ln(n) && \text{since} \ln(n) \text{ is an increasing function and } n \geq n_0 = 1 \\
&= n \cdot \ln(n) \\
&= 1 \cdot n \cdot \ln(n) \\
&= c \cdot n\ln(n)
\end{aligned}
$$

therefore $\exists c \in \mathbb{R}^+, \exists n_0 \in \mathbb{N}, \forall n \geq n_0, \ln(n!) \leq c \cdot n\ln(n)$

which is the definition of $\ln(n!) \in O(n \ln n)$

3b

proving the hint first:
$\forall n \in \mathbb{N}, n^n \geq (n+1)^{n-1}$

base case $n = 1$

$$1^1 \geq (1+1)^{1-1}$$
$$1 \geq 2^0$$
$$1 \geq 1$$

Assume for inductive hypothesis

$$k^k \geq (k+1)^{k-1}$$

for $1 \leq k$ and $k \in \mathbb{N}$

from the inductive hypothesis

$$
\begin{aligned}
k^k &\geq (k+1)^{k-1} \\
k^k &\geq \frac{(k+1)^k}{k+1} \\
\frac{k^k}{(k+1)^k} &\geq \frac{1}{k+1} \\
\left(\frac{k}{k+1}\right)^k &\geq \frac{1}{k+1} \\
\left(\frac{k+1}{k+2}\right)^k > \left(\frac{k}{k+1}\right)^k &\geq \frac{1}{k+1} \ (*) \\
\left(\frac{k+1}{k+2}\right)^k &\geq \frac{1}{k+1} \\
\frac{(k+1)^k}{(k+2)^k} &\geq \frac{1}{k+1} \\
(k+1)^k &\geq \left(\frac{(k+2)^k}{k+1}\right) \\
(k+1)^k(k+1) &\geq (k+2)^k \\
(k+1)^{k+1} &\geq (k+2)^k \\
(k+1)^{k+1} &\geq (k+1+1)^{k+1-1}
\end{aligned}
$$

to explain (*), if we put $\frac{k+1}{k+2}$ and $\frac{k}{k+1}$ into a ratio, we end up with

$$\frac{k^2+2k+1}{k^2+2k}$$

show casing that $\frac{k+1}{k+2}$ is strictly greater than $\frac{k}{k+1}$

therefore by mathematical induction, we have

$$\forall n \in \mathbb{N}, n^n \geq (n+1)^{n-1}$$

Next, proving $(n!)^2 \geq n^n, \forall n \in \mathbb{N}$

base case, $n = 1$

$$(1!)^2 = 1 \geq 1 = 1^1$$

For inductive hypothesis assume that $(k!)^2 \geq k^k, \forall k \in \mathbb{N}, k \geq 1$

$$
\begin{aligned}
((k+1)!)^2 &= ((k+1)k!)^2 \\
&= (k+1)^2(k!)^2 \\
&\geq (k+1)^2 \cdot k^k && \text{by the inductive hypothesis} \\
&= (k+1) \cdot (k+1) \cdot k^k \\
&\geq (k+1) \cdot (k+1) \cdot (k+1)^{k-1} && \text{by using the hint} \\
&= (k+1) \cdot (k+1)^k \\
&= (k+1)^{k+1}
\end{aligned}
$$

therefore by mathematical induction we have

$$(n!)^2 \geq n^n, \forall n \in \mathbb{N}$$

■

3c

from part a, we have $\ln(n!) \in O(n\ln(n))$

we now seek to prove $\ln(n!) \in \Omega(n\ln(n))$

choose $c = \frac{1}{2}$ and $n_0 = 1$,

$$\begin{aligned}
(n!)^2 &\geq n^n && \text{from part b} \\
\ln(n!^2) &\geq \ln(n^n) && \text{since } \ln \text{ is a increasing function and } n \geq n_0 \geq 1 \\
2\ln(n!) &\geq \ln(n^n) \\
\ln(n!) &\geq \frac{1}{2}\ln(n^n) && \text{divide both sides by 2}
\end{aligned}$$

then we get

$$\begin{aligned}
\ln(n!) &\geq \frac{1}{2}n\ln(n) \\
&= cn\ln(n)
\end{aligned}$$

therefore $\exists c \in \mathbb{R}^+, \exists n_0 \in \mathbb{N}, \forall n \geq n_0, c \cdot n\ln(n) \leq \ln(n!)$

which is the definition of $\ln(n!) \in \Omega(n\ln(n))$

and since we have both $\ln(n!) \in \Omega(n\ln(n))$ and $\ln(n!) \in O(n\ln(n))$

we can conclude $\ln(n!) \in \Theta(n\ln(n))$

Question 4

`H` is home `A` is school A `B` is school B `W` is work

a)

3 times for symmetric

6 times for asymmetric

b)

2 times for symmetric, because running from (home, A) gives implicitly (home, B), and we just need to run it again from (A,B)

3 times for asymmetric.

we need all 6,

```
(H, A)
(A, H)
(A, B)
(B, A)
(H, B)
(B, H)
```

running noodle maps on `(H, A)` gives also `(H, B)`

again, on `(A, B)` gives `(A, H)`

again on `(B, H)` gives `(B, A)`

thus, getting all 6 in 3 runs

c)

if We are working only under part a with part c,

5 times if symmetric.

```
(H, A)
(H, B)
(A, B)
(A, W)
(B, W)
```

if asymmetric, it's 6 times, one per each edge

```
(H, A)
(H, B)
(A, B)
(B, A)
(A, W)
(B, W)
```

now working under part b

2 times for symmetric. in this case we need four edges,

```
(H, A)
(B, A)
(B, W)
(A, W)
```

run it on `(B, A)` gives `(B, W)`

run it again on `(A, W)` gives `(H, A)` as well.

in the asymmetric case, it's 3 times

```
(H, A)
(H, B)

(A, W)
(A, B)

(B, W)
(B, A)
```

run it on `(A, W)` gives `(A, B)`

run it on `(B, A)` gives, `(B, W)`

run it on `(H, A)` gives `(H, B)`